

Vinh Ly

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Summary

Software Engineer with **18 years** building large scale backend systems at Microsoft, Google, YouTube, and Whatnot. Owner of **Go**, **Python**, **C++**, **Kubernetes**, and **Bigtable** services across ads, video infrastructure, and live commerce. Shipped systems serving billions of daily requests with 99.99% availability on call. Partnered with ML, infra, and product teams to cut tail latency 40% in core ranking pipelines. Next focus: petabyte scale data ingestion and storage layers powering AI/ML experiment tracking. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Go, Python, C++, Java, TypeScript, JavaScript, HTML, CSS

Data and Storage: Bigtable, MySQL, Postgres, Kafka, PubSub

Infrastructure: Kubernetes, Docker, Terraform, GCP, AWS

Distributed Systems: gRPC, Spanner, Flume, Dremel, Protobuf

Reliability Practices: incident response, capacity planning, load testing

Work Experience

Staff Software Engineer, Whatnot

05/2024 - 02/2025 | Remote

- Owned backend services in **Go** and **Python** for live commerce checkout and auction flows processing millions of concurrent bids per minute.
- Migrated live state stores from **Postgres** to **Bigtable** on **GCP**, cutting storage cost 60% and p99 read latency 35%.
- Designed **Kafka** backed event pipelines on **Kubernetes** that reduced order processing tail latency from 800ms to 120ms.
- Led on call rotation for 6 production services with strict **SLOs** and incident response playbooks.
- Partnered with trust and safety to ship anti-fraud signal ingestion handling 50K events per second.
- Built observability tooling using **OpenTelemetry** and **Prometheus** to surface hot keys across auction shards.

Staff Software Engineer, YouTube

02/2025 - Present | Remote

- Architecting **C++** and **Go** services for video metadata indexing across petabyte scale datasets on **Bigtable** and **Spanner**.
- Reduced video upload ingest end to end latency by 30% by redesigning **PubSub** fanout topology.
- Own end to end migration of legacy **Java** services to **gRPC** with **Protobuf** schemas.
- Drive capacity planning for storage clusters exceeding 100PB of daily active metadata.
- Mentor 4 senior engineers on distributed systems design and production readiness reviews.

Staff Software Engineer/Manager, Google

03/2015 - 04/2024 | Remote

- Managed a team of 7 engineers building ads ranking infrastructure in **C++** and **Java** serving billions of queries daily.
- Drove migration from monolithic **MySQL** shards to **Spanner** across 3 global regions with consistent cross region replication.
- Shipped real time bidding pipelines on **Flume** and **PubSub** with 99.99% availability measured across rolling windows.
- Partnered with ML platform to expose feature stores behind stable **gRPC** contracts used in production ranking.
- Owned **Dremel** query layer optimizations that cut interactive analytics scan time 45% on multi-petabyte tables.

Software Engineer, Google

02/2011 - 02/2015 | Remote

- Owned on call for 4 critical services, drove mean time to recovery from 45 minutes to 12 minutes.
- Improved ranking cache hit rate from 78% to 92% by introducing tiered **Bigtable** lookups and predictive prefetch.
- Built core components of the ads serving stack in **C++** handling billions of requests per day.
- Designed and rolled out **Protobuf** schema versioning used by 30+ downstream consumers.

Software Design Engineer II, Microsoft

10/2007 - 12/2011 | Remote

- Reduced network stack memory usage by 18% through buffer pooling redesign across the transport layer.
- Developed Windows networking components in **C++** shipping in Windows 7 and Windows Server releases.
- Owned kernel level drivers for IPv6 transition protocols with strict reliability requirements.
- Contributed to internal **SQL Server** tooling in **C#** for production diagnostics.

Software Design Engineer Intern, Microsoft

06/2006 - 08/2006 | Remote

- Authored test cases that surfaced 3 latent bugs in the network driver validation suite.
- Implemented internal utilities in **C++** for the Windows kernel test harness.

Education

B.S. Honors degrees in Computer Science

Virginia Tech | 2004 - 2007